

A Prior-Information-Based Unbiased Convex Estimator for the Classical Linear Regression Model

Mustafa I. Alheety¹, HM Nayem², B. M. Golam Kibria²

¹Department of Mathematics, College of education for pure sciences, University of Anbar
Ramadi, Al Anbar 31001, Iraq

²Department of Mathematics and Statistics, Florida International University, Miami, FL
33199

Abstract

This study introduces a novel unbiased restricted estimator that effectively integrates prior information into the estimation framework. A comparative assessment with several existing estimators demonstrates clear analytical advantages, particularly in terms of bias reduction and mean squared error performance. The simulation results consistently reveal that the proposed estimator yields superior performance across varying parameter settings and sample sizes. Collectively, the findings underscore the estimator's potential as a powerful alternative to conventional approaches, offering both methodological novelty and practical improvements in statistical inference.

Keywords: Prior-Information, Unbiased, Multicollinearity, Convex, Restricted

1 Introduction

In the standard linear regression framework, the ordinary least squares (OLS) estimator is widely used due to its unbiasedness and optimality under the Gauss–Markov assumptions (Akdeniz and Erol, 2003; Nayem et al., 2024). However, one of the critical assumptions is the absence of strong linear dependence among explanatory variables. When multicollinearity is present, the design matrix becomes ill-conditioned, leading to inflated variances of the estimated coefficients (Hoque and Kibria, 2023). This not only reduces the precision of estimates but also undermines the reliability of statistical inference, thereby complicating decisions regarding hypothesis testing (Yasmin and Kibria, 2025).

To overcome this limitation, several biased estimation techniques have been introduced (Kibria and Banik, 2016). Ridge regression, proposed by Hoerl and Kennard (1970), is among the most prominent approaches, as it stabilizes estimation by imposing a penalty on large coefficients (Hoerl and Kennard, 1970). Despite its usefulness, ridge regression remains biased, and evaluation based solely on variance reduction is inadequate. Crose et al., (1995) addressed this issue by introducing the unbiased ridge regression (URR) estimator, which retains the advantages of ridge regression while improving upon mean squared error (MSE) performance compared to both OLS and ordinary ridge regression (ORR) estimator (Crose et al., 1995).

Another important direction involves incorporating prior or external information into estimation. The restricted least squares (RLS) estimator achieves lower variance than OLS by exploiting such restrictions. Building on this, Sarkar (1992) proposed a restricted ridge regression estimator, combining regularization with parameter restrictions (Sarkar, 1992). Recent literature also highlights that convex combinations of estimators can inherit complementary strengths, yielding estimators with superior statistical properties, particularly in terms of efficiency and MSE reduction.

Motivated by these developments, this article introduces an unbiased estimator that integrates prior information with the restricted ridge regression framework using a convex combination technique. The proposed approach not only alleviates the adverse effects of multicollinearity but also provides more accurate and efficient parameter estimates than existing alternatives, making it a valuable tool in applied settings.

2 The Proposed Convex Estimator

In the presence of multicollinearity, the OLS estimator loses efficiency, as correlations among explanatory variables inflate the variance of the estimates, increasing with the severity of dependence. For this reason, Hoerl and Kennard (1970) proposed an estimator that will reduce the inflation of the variance as follows:

$$\hat{\beta}(k) = (X'X + kI)^{-1}X'Y, \quad k \geq 0$$

They called it ordinary ridge regression estimator (ORR). Crouse et al. (1995) introduced the unbiased ridge estimator (URR) as follows:

$$\hat{\beta}(k) = (X'X + kI)^{-1}(X'Y + kJ), \quad k \geq 0$$

where J is a random vector with $J \sim N(\beta, (\sigma^2/k)I)$ and independent of $\hat{\beta}$.

The restricted least squares (RLS) method enables the use of sample information and prior information simultaneously. The RLS estimator can be written as follows:

$$\hat{\beta}_{RLS} = \hat{\beta} + S^{-1}R'(RS^{-1}R')^{-1}(r - R\hat{\beta})$$

The convex estimator can be given as follows:

$$\hat{\beta}_{con} = A\tilde{\beta}_1 + (I - A)\tilde{\beta}_2$$

where $\tilde{\beta}_1$ and $\tilde{\beta}_2$ be any two estimators and A is a square matrix. The convex estimators have the ability to take advantage of the properties of the estimators that compose them and have the potential to deal with multicollinearity effectively.

Wu (2014) studied the philosophy of prior information on β as follows:

$$\hat{\beta}(C, J) = C\hat{\beta} + (I - C)J$$

where C represented $p \times p$ matrix J is a random variable distributed as normal with mean β and variance-covariance matrix V , $V = \sigma^2(I - C)^{-1}CS^{-1}$. The unbiased estimator was obtained by taking in account the Two-Parameter estimator proposed by Ozkale and Kaciraular (2007). So, the goal in this paper is to present a generalization of above equation by replacing $\hat{\beta}_{RLS}$ instead of $\hat{\beta}$ to get a new family of unbiased restricted convex estimator.

In fact, the true parameter β may has prior information as a vector called J . Also, if we assume the linear restrictions $R\beta = r$ are satisfied in the model, we can suggest the following general form of convex estimator as follows:

$$\hat{\beta}_{CE} = C\hat{\beta}_{RLS} + (I - C)J$$

To study the optimality of $\hat{\beta}_{CE}$, we find the form of the matrix C that makes the mean squared error matrix (MMSE) of $\hat{\beta}_{CE}$ minimum.

For that, the MMSE of $\hat{\beta}_{CE}$ is derived as follows:

$$L = \text{MMSE}(\hat{\beta}_{CE}) = C \text{Var}(\hat{\beta}_{RLS}) C' + (I - C)V(I - C)'$$

A minimum will be existing when L is a real-valued, convex and differentiable function. Therefore, by differentiating the above equation with respect to C and equating to zero, we obtain:

$$\frac{\partial L}{\partial C} = 2C \text{Var}(\hat{\beta}_{RLS}) - 2(I - C)V = 0$$

And then

$$C = V(\text{Var}(\hat{\beta}_{RLS}) + V)^{-1}$$

Therefore,

$$\begin{aligned} V &= (I - C)^{-1} C \text{Var}(\hat{\beta}_{RLS}) \\ &= \sigma^2 (I - C)^{-1} C M S^{-1} \end{aligned}$$

where $\text{Var}(\hat{\beta}_{RLS}) = \sigma^2 M S^{-1}$ and $M = I - S^{-1}R'(R S^{-1} R')^{-1} R$.

So, the proposed new estimator will be:

$$\hat{\beta}_{CE} = W\hat{\beta}_{RLS} + (I - W)J$$

where $W = (I + k S^{-1})^{-1}$, $k \geq 0$ and $V = \sigma^2 (I - W)^{-1} W M S^{-1}$. The random variable $J \sim N(\beta, \sigma^2 (I - W)^{-1} W M S^{-1})$ and we call it the Unbiased Convex (UC) estimator. The UC estimator is unbiased since:

$$E(\hat{\beta}_{UC}) = W\beta + (I - W)\beta = \beta.$$

Also the variance-covariance matrix of UC estimator is given as follows:

$$\text{Cov}(\hat{\beta}_{UC}) = \sigma^2 W M S^{-1}$$

3. Simulation Study

3.1 Simulation Technique

For a numerical comparison among the estimators, a simulation study has been conducted in this section. It consists of two parts: Simulation Technique and simulation result discussion. The data generation process for the models was carried out according to a widely used method as described below (Kibria, 2003; Nayem and Kibria, 2024).

$$x_{ij} = (1 - \rho^2)^{1/2} z_{ij} + \rho z_{i,p+1}, i = 1, 2, \dots, n \text{ and } j = 1, 2, \dots, p$$

where ρ denotes the correlation coefficient between any two explanatory variables, z_{ij} denotes the independent pseudo-random variable obtained from standard normal distribution and p denotes the number of explanatory variables. Moreover, the dependent variable Y was obtained through the following equation (Alheety et al., 2025).

$$y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \cdots + \beta_p x_{ip} + \varepsilon_i, i = 1, 2, \dots, n.$$

where, ε_i is the error term, i.e. $\varepsilon_i \sim N(0, \sigma^2)$.

In simulation, two different values of explanatory variables, $p = 5$ and 10 , two different values of the correlation coefficients between the explanatory variables, $\rho = 0.9$ and 0.95 , three different values of the sample sizes, $n = 20, 50$, and 100 , and the error variance, $\sigma^2 = 1$. The data generation process of the explanatory variables was carried out with the help of the values determined for p , ρ , n , and σ . In this study, we employed the ridge parameter, denoted as k , within a specified range from 0.1 to 0.9 . This range was explored in increments of 0.1 , allowing for a comprehensive examination of how varying levels of k influence the outcomes of our analysis. The experiment was repeated, $N = 10,000$ times. The following restrictions were set for different predictors.

When $p = 5$,

$$R = \begin{bmatrix} 1 & 1 & -2 & 0 & 0 \end{bmatrix} \text{ and } r = \begin{bmatrix} 0 \end{bmatrix}$$

When $p = 10$,

$$R = \begin{bmatrix} 1 & 1 & 0 & 0 & 1 & 0 & -2 & -1 & 1 & 0 \end{bmatrix} \text{ and } r = \begin{bmatrix} 0 \end{bmatrix}$$

The MSE was computed by evaluating the estimated coefficients of each model against the true coefficients (assumed $\beta = I$). For any estimator $\hat{\beta}$ of β , the MSE can be measured by (Hoque and Kibria, 2025).

$$\text{MSE}(\hat{\beta}, \beta) = E(\hat{\beta} - \beta)'(\hat{\beta} - \beta)$$

The averaged MSE was calculated based on the following formula over all simulations (Nayem et al., 2025).

$$\text{AMSE}(\hat{\beta}_i^*) = \frac{1}{N} \sum_{i=1}^P (\hat{\beta}_i - \beta)'(\hat{\beta}_i - \beta)$$

where $\hat{\beta}_i^*$ is any of the URR, RLS, UC, ORR, or OLS estimators, β is the true parameter and p is the number of predictors.

3.2 Simulation Results Discussion

3.2.1 Performance as a Function of n

With only 20 samples, the UC estimator tends to exhibit a higher MSE, especially when the predictors are highly correlated. While UC estimator still performs relatively well with limited samples compared to other estimators, significant correlations such as 0.90, among predictors contribute to increased multicollinearity, subsequently raising the MSE (as illustrated in Tables 1, and 4). This indicates that with smaller sample sizes, the negative effects of multicollinearity on estimator performance are more pronounced.

As the sample size increases from 20 to 50, there is a marked reduction in MSE for the UC estimator. The added data provides a stabilizing effect, enhancing the estimator's resilience to the multicollinearity caused by high correlations. This improvement in MSE at moderate sample sizes underscores that although high predictor correlation remains a challenge, a larger number of samples helps mitigate its impact on the UC estimator's accuracy.

At a sample size of 100, the UC estimator consistently achieves its lowest MSE values, even under extreme predictor correlations (e.g., 0.99) (Tables 3 and 6). With larger samples, the UC estimator displays robustness, as the effect of high correlation among predictors on the MSE diminishes. This

trend illustrates the advantages of having more data, as the UC estimator effectively reduces the influence of multicollinearity and maintains a low MSE under challenging high-correlation conditions and multiple predictors.

Table 1 Average MSE values of the estimators when $p = 5$ and $\rho = 0.90$

k	n=20					n=50					n=100				
	URR	RLS	UC	ORR	OLS	URR	RLS	UC	ORR	OLS	URR	RLS	UC	ORR	OLS
0.1	9.91	8.28	7.42	9.90	11.18	4.43	3.64	3.55	4.43	4.55	2.72	2.31	2.29	2.72	2.74
0.2	8.89	8.18	6.69	8.87	11.09	4.33	3.65	3.48	4.33	4.57	2.66	2.30	2.26	2.66	2.71
0.3	8.12	8.23	6.20	8.10	11.12	4.18	3.62	3.37	4.18	4.53	2.65	2.30	2.24	2.65	2.73
0.4	7.44	8.22	5.74	7.41	11.05	4.07	3.61	3.28	4.06	4.51	2.63	2.30	2.22	2.64	2.74
0.5	6.86	8.17	5.32	6.84	11.05	4.00	3.65	3.25	4.00	4.56	2.58	2.29	2.19	2.58	2.71
0.6	6.47	8.20	5.02	6.43	11.14	3.92	3.66	3.18	3.92	4.57	2.57	2.30	2.19	2.57	2.72
0.7	6.10	8.37	4.81	6.05	11.19	3.82	3.65	3.11	3.82	4.55	2.55	2.29	2.16	2.55	2.73
0.8	5.69	8.23	4.50	5.65	11.07	3.73	3.62	3.03	3.72	4.54	2.51	2.28	2.14	2.51	2.71
0.9	5.40	8.24	4.30	5.35	11.12	3.65	3.63	2.98	3.64	4.53	2.51	2.29	2.12	2.51	2.73

Table 2 Average MSE values of the estimators when $p = 5$ and $\rho = 0.95$

k	n=20					n=50					n=100				
	URR	RLS	UC	ORR	OLS	URR	RLS	UC	ORR	OLS	URR	RLS	UC	ORR	OLS
0.1	15.67	14.03	11.55	15.65	19.37	6.93	5.65	5.38	6.93	7.31	3.97	3.28	3.22	3.97	4.05
0.2	13.10	13.93	9.82	13.07	19.16	6.65	5.71	5.18	6.64	7.38	3.86	3.25	3.13	3.86	4.02
0.3	11.35	13.96	8.61	11.28	19.14	6.35	5.72	4.96	6.34	7.39	3.82	3.28	3.10	3.81	4.06
0.4	10.09	14.01	7.68	9.99	19.26	6.00	5.69	4.73	5.99	7.32	3.75	3.28	3.04	3.74	4.06
0.5	9.08	14.26	7.00	8.98	19.56	5.78	5.69	4.54	5.76	7.37	3.66	3.27	2.98	3.66	4.05
0.6	8.24	14.17	6.36	8.12	19.65	5.56	5.72	4.39	5.54	7.38	3.60	3.30	2.96	3.60	4.05
0.7	7.48	14.25	5.86	7.35	19.44	5.31	5.71	4.22	5.29	7.35	3.57	3.31	2.92	3.56	4.09
0.8	6.81	13.87	5.33	6.67	19.23	5.07	5.63	4.02	5.05	7.29	3.48	3.30	2.87	3.48	4.06

3.2.2 Performance as a Function of the number of predictors, p

When the predictor set comprises a small number of variables ($p = 5$), the UC estimator consistently outperforms its counterparts, particularly in terms of MSE. However, despite these complexities, the UC estimator retains a competitive advantage, effectively reducing MSE more efficiently than its peers as the sample size expands, demonstrating its robust performance in managing more intricate models (Tables 1, 2, and 3).

The challenges intensify with ten predictors ($p = 10$), where the detrimental effects of multicollinearity and an expanded parameter space are exacerbated. In situations marked by high correlation (e.g., 0.95 and 0.99), we observe a notable increase in MSE due to the inherent difficulties in accurate parameter estimation. Nonetheless, the UC estimator exhibits resilience in these challenging conditions. As sample sizes reach 50 or 100, the performance of UC shows marked improvement, underscoring its adaptability in navigating complex interrelationships among multiple predictors (Tables 5 and 6).

3.2.3 Performance as a Function of Correlation Coefficients, ρ

The correlation levels among the predictors, specifically 0.90, 0.95, and 0.99, have a significant impact on the performance of the various estimators. This is clearly demonstrated in the analyses presented in Tables 1 through 6. Generally, higher correlation values are associated with an increase in MSE, indicating a decline in estimation accuracy, as shown in Tables 1 through 6.

At a correlation level of 0.90, multicollinearity is evident but remains manageable. Notably, the UC estimator consistently demonstrates superior performance, as indicated by its relatively low MSE values, particularly when examining larger sample sizes. In instances where sample sizes are increased, such as 50 or 100, the UC estimator continues to yield reliable estimates despite the prevailing multicollinearity (Table 1 and 4).

As correlation levels rise to 0.95, the adverse effects of multicollinearity become increasingly pronounced, leading to a marked escalation in MSE across all estimators. Nevertheless, the UC estimator maintains a competitive advantage; it experiences a comparatively smaller increase in MSE relative to traditional estimators. The detrimental impacts of heightened correlation are especially observable in smaller samples, where the amplified collinearity exacerbates challenges for the estimators. Notably, as sample sizes expand, the UC estimator reveals its robustness by effectively minimizing MSE in comparison with alternative estimators (Tables 2 and 5).

At the maximum correlation level of 0.99, the severity of multicollinearity poses significant obstacles to estimation accuracy. All estimators are subject to a pronounced increase in MSE, with the effects being severe in smaller sample sizes. In these circumstances, the UC estimator displays remarkable resilience, consistently yielding lower MSE values compared to its counterparts. This advantage becomes particularly salient as the sample size approaches 50 or 100, wherein the UC estimator continues to furnish more stable and accurate estimates (Tables 3 and 6).

Table 3 Average MSE values of the estimators when $p = 5$ and $\rho = 0.99$

k	n=20					n=50					n=100				
	URR	RLS	UC	ORR	OLS	URR	RLS	UC	ORR	OLS	URR	RLS	UC	ORR	OLS
0.1	37.71	56.31	28.02	37.47	79.22	21.76	20.72	16.35	21.73	27.80	12.22	10.32	9.29	12.21	13.63
0.2	23.76	56.06	17.95	23.39	79.34	17.75	20.76	13.45	17.68	27.81	11.06	10.41	8.50	11.04	13.63
0.3	16.81	56.07	12.96	16.46	78.65	14.73	20.68	11.26	14.62	27.62	10.12	10.45	7.79	10.08	13.71
0.4	13.09	56.49	10.07	12.67	80.06	12.71	20.76	9.71	12.56	27.81	9.25	10.34	7.11	9.20	13.64
0.5	10.57	56.84	8.27	10.12	78.87	10.97	20.74	8.45	10.80	27.63	8.60	10.51	6.67	8.54	13.80
0.6	8.63	55.85	6.76	8.22	79.37	9.64	20.55	7.44	9.46	27.60	7.99	10.53	6.20	7.91	13.81
0.7	7.37	56.05	5.84	6.94	79.65	8.63	20.70	6.66	8.41	27.76	7.43	10.49	5.78	7.36	13.82
0.8	6.43	55.60	5.08	5.97	79.68	7.72	20.50	6.01	7.53	27.60	6.92	10.47	5.42	6.82	13.73
0.9	5.66	56.29	4.51	5.23	79.31	7.03	20.65	5.49	6.81	27.75	6.51	10.54	5.11	6.42	13.81

Table 4 Average MSE values of the estimators when $p = 10$ and $\rho = 0.90$

k	n=20					n=50					n=100				
	URR	RLS	UC	ORR	OLS	URR	RLS	UC	ORR	OLS	URR	RLS	UC	ORR	OLS
0.1	45.16	49.69	38.39	45.14	60.04	16.49	15.17	14.56	16.49	17.22	8.37	7.65	7.53	8.37	8.50
0.2	37.14	49.50	31.88	37.08	59.78	15.90	15.26	14.07	15.90	17.29	8.17	7.54	7.31	8.17	8.42
0.3	31.87	49.80	27.70	31.81	60.39	15.18	15.14	13.43	15.18	17.17	8.12	7.63	7.29	8.12	8.50
0.4	27.91	49.59	24.41	27.81	60.10	14.73	15.22	13.02	14.73	17.31	8.02	7.66	7.22	8.02	8.52
0.5	24.79	49.53	21.82	24.70	60.39	14.13	15.19	12.54	14.12	17.21	7.94	7.68	7.13	7.94	8.57
0.6	22.41	49.53	19.89	22.31	60.14	13.64	15.19	12.13	13.63	17.20	7.76	7.60	6.96	7.76	8.50
0.7	20.62	49.95	18.30	20.51	60.72	13.24	15.26	11.79	13.22	17.26	7.63	7.61	6.87	7.63	8.47
0.8	19.00	49.65	16.95	18.86	60.16	12.80	15.22	11.37	12.78	17.28	7.54	7.61	6.78	7.54	8.49
0.9	17.42	49.43	15.60	17.28	60.12	12.33	15.12	10.96	12.31	17.19	7.43	7.60	6.68	7.43	8.49

Table 5 Average MSE values of the estimators when $p = 10$ and $\rho = 0.95$

k	n=20					n=50					n=100				
	URR	RLS	UC	ORR	OLS	URR	RLS	UC	ORR	OLS	URR	RLS	UC	ORR	OLS

0.1	72.66	93.48	61.97	72.60	113.87	28.69	27.29	25.32	28.69	30.97	14.31	13.13	12.76	14.31	14.73
0.2	55.13	93.42	47.63	55.05	114.63	26.64	26.99	23.34	26.63	30.93	13.98	13.19	12.46	13.98	14.81
0.3	45.21	94.24	39.67	45.04	114.83	25.10	27.23	22.08	25.08	31.15	13.55	13.16	12.10	13.55	14.76
0.4	37.53	92.29	33.16	37.38	112.89	23.46	27.19	20.72	23.44	31.02	13.23	13.16	11.79	13.22	14.81
0.5	32.88	94.28	29.33	32.71	115.22	22.17	27.26	19.60	22.15	31.12	12.84	13.10	11.43	12.83	14.75
0.6	28.52	93.18	25.48	28.35	114.01	20.97	27.29	18.53	20.95	31.23	12.53	13.18	11.20	12.52	14.78
0.7	25.53	93.57	22.89	25.30	114.33	19.91	27.49	17.68	19.87	31.31	12.13	13.08	10.85	12.13	14.69
0.8	22.92	93.23	20.65	22.71	114.28	18.82	27.38	16.73	18.78	31.20	11.90	13.12	10.61	11.89	14.78
0.9	20.74	92.92	18.76	20.50	113.72	18.01	27.39	15.98	17.97	31.34	11.63	13.13	10.38	11.62	14.78

Table 6 Average MSE values of the estimators when $p = 10$ and $\rho = 0.99$

k	n=20					n=50					n=100				
	URR	RLS	UC	ORR	OLS	URR	RLS	UC	ORR	OLS	URR	RLS	UC	ORR	OLS
0.1	154.64	411.91	135.42	154.22	509.28	95.73	113.37	83.70	95.65	130.69	51.82	51.80	45.59	51.81	58.98
0.2	90.09	408.52	80.64	89.56	510.27	74.54	113.33	65.50	74.42	130.68	46.51	52.38	41.00	46.48	59.63
0.3	62.19	413.49	56.19	61.64	516.98	60.80	113.92	53.44	60.64	131.92	41.69	52.52	36.87	41.63	59.61
0.4	46.27	411.57	42.12	45.74	514.11	50.52	114.09	44.85	50.36	131.27	37.45	52.49	33.30	37.42	59.39
0.5	36.45	413.55	33.23	35.89	512.91	43.05	114.72	38.36	42.85	131.61	33.71	51.84	29.93	33.63	58.72
0.6	29.30	410.94	26.83	28.74	509.52	37.04	113.75	32.96	36.82	131.22	31.16	52.31	27.63	31.07	59.46
0.7	24.53	414.69	22.47	23.95	517.42	32.16	113.15	28.70	31.91	130.24	28.41	52.16	25.26	28.32	59.11
0.8	20.74	420.51	19.07	20.18	523.58	28.39	113.29	25.38	28.17	130.45	26.59	52.77	23.62	26.48	59.93
0.9	17.95	414.00	16.45	17.39	511.75	25.51	113.29	22.82	25.22	130.27	24.55	52.46	21.81	24.43	59.68

4. Conclusion

This study introduces a robust, unbiased restricted estimator that seamlessly integrates prior information into the classical linear regression framework, offering significant improvements over traditional estimation methods. The estimator's unique ability to incorporate prior knowledge not only addresses limitations seen in conventional estimators, but also enhances the precision of parameter estimates, particularly when prior information is available or when working with small sample sizes.

The comparative analysis with a range of widely used estimators highlights the clear advantages of the proposed method in terms of MSE performance. The estimator consistently outperforms its counterparts under varying simulation scenarios, showcasing its flexibility and robustness across diverse parametric conditions. Furthermore, the study's comprehensive simulation results demonstrate that the proposed estimator excels in scenarios characterized by different sample sizes, model specifications, and levels of noise, making it an ideal choice for a variety of real-world applications.

In conclusion, this paper presents a novel estimator that holds significant promise for advancing the field of linear regression modeling. By leveraging prior information, this approach provides a powerful tool for researchers, offering an efficient and unbiased alternative to conventional estimation techniques. The findings from simulation studies suggest that this estimator should be seriously considered by statisticians and data analysts, particularly when the integration of prior knowledge is crucial for enhancing the accuracy of statistical predictions.

Acknowledgments

The authors wish to acknowledge the contributions of previous researchers whose foundational work has significantly informed and guided this study.

References

- Akdeniz, F., Erol, H. (2003). Mean squared error matrix comparisons of some biased estimators in linear regression. *Communications in Statistics—Theory and Methods* 32:2389–2413.
- Alheety, M. I., Nayem, H. M., & Kibria, B. G. (2025). An unbiased convex estimator depending on prior information for the classical linear regression model. *Stats*, 8(1), 16.
- Crouse RH, Jin C, Hanumara RC (1995) Unbiased ridge estimator with prior information and ridge trace. *Communications in Statistics—Theory and Methods* 24:2341–2354.
- Hoerl, A. E., Kennard, R. W. (1970). Ridge regression: biased estimation for non-orthogonal problems. *Technometrics* 12:69–82.
- Hoque, M. A., & Kibria, B. M. (2023). Some one and two parameter estimators for the multicollinear Gaussian linear regression model: Simulations and applications. *Surveys in Mathematics and its Applications*, 18, 183-221.
- Hoque, M. A., Bursac, Z., & Kibria, B. G. (2025). Inferences About Two-Parameter Multicollinear Gaussian Linear Regression Models: An Empirical Type I Error and Power Comparison. *Stats*, 8(2), 28.
- Kibria, B. M. G., and Banik S., (2016), Some ridge regression estimators and their performance, *Journal of Modern Applied Statistical Methods*. 15, 206-238.
- Kibria, B. M. G. (2003). Performance of Some New Ridge Regression Estimators. *Communications in Statistics - Simulation and Computation*, 32(2), 419–435. <https://doi.org/10.1081/SAC-120017499>

- Najarian, S., Arashi, M., & Kibria, B. G. (2012). A simulation study on some restricted ridge regression estimators. *Communications in Statistics-Simulation and Computation*, 42(4), 871-890.
- Nayem, H. M., Aziz, S., & Kibria, B. G. (2024). Comparison among ordinary least squares, ridge, lasso, and elastic net estimators in the presence of outliers: Simulation and application. *International Journal of Statistical Sciences*, 25-48.
- Nayem, H. M., Aziz, S., & Kibria, B. G. (2025). Evaluating Estimator Performance Under Multicollinearity: A Trade-Off Between MSE and Accuracy in Logistic, Lasso, Elastic Net, and Ridge Regression with Varying Penalty Parameters. *Stats*, 8(2), 45.
- Özkale, M. R., & Kaciranlar, S. (2007). The restricted and unrestricted two-parameter estimators. *Communications in Statistics—Theory and Methods*, 36(15), 2707-2725.
- Sarkar, N. 1992 . “ A new estimator combining the ridge regression and the restricted least squares method of estimation ” . *Communications in Statistics—Theory and Methods* 21(7), 1987 – 2000.
- Wu, Jibo (2014). “An unbiased two- parameter estimator with prior information in linear regression model ”. *The Scientific Word Journal* Volume (2014), 8 pages.
- Yasmin, N., & Kibria, B. G. (2025). Performance of Some Improved Estimators and their Robust Versions in Presence of Multicollinearity and Outliers: N. Yasmin and BMG Kibria. *Sankhya B*, 87(1), 173-219.